QuickPhone — Physical ERD (complete): 16 tables with all keys and attributes

One diagram for the whole data model. Each line is named both ways: top or left from the "one" side, bottom or right back from the "many" side. PK primary key · FK foreign key (→ table) · UK unique · CK checked value list · NN not null · derived = denormalized

store — D7 Staff & Stores

PK	**store_id**	INTEGER	NN
UK	store_name	VARCHAR(60)	NN
	address	VARCHAR(200)	NN
	phone	VARCHAR(20)	NN
	email_address	VARCHAR(120)	NN
	business_hours	VARCHAR(120)	NN

appointment — D2 Appointments

PK	**appointment_id**	INTEGER	NN
FK	customer_id	INTEGER	→ customer
FK	store_id	INTEGER	→ store
FK	technician_id	INTEGER	→ employee
	scheduled_at	TIMESTAMP	NN
CK	appt_status	VARCHAR(12)	NN

repair_job — D3 Repair Jobs

PK	**repair_job_id**	INTEGER	NN
UK	ticket_number	VARCHAR(12)	NN
FK	device_id	INTEGER	→ device
FK	store_id	INTEGER	→ store
FK,UK	appointment_id	INTEGER	→ appointment
FK	assigned_tech_id	INTEGER	→ employee
CK	status	VARCHAR(20)	derived · NN
	reported_issue	VARCHAR(500)	NN
	device_passcode	VARCHAR(255)	encrypted
	intake_condition	VARCHAR(500)	
	diagnosis_notes	VARCHAR(2000)	
	repair_notes	VARCHAR(2000)	
	check_in_at	TIMESTAMP	NN
	completed_at	TIMESTAMP	
	final_cost	DECIMAL(10,2)	
	email_sent_at	TIMESTAMP	
FK	email_sent_by_id	INTEGER	→ employee
CK	email_status	VARCHAR(10)	
	picked_up_at	TIMESTAMP	
	id_verified_flag	BOOLEAN	NN
	signature_file	VARCHAR(255)	

payment — D3 Repair Jobs

PK	**payment_id**	INTEGER	NN
FK	repair_job_id	INTEGER	→ repair_job
	amount	DECIMAL(10,2)	NN
CK	method	VARCHAR(12)	NN
CK	payment_status	VARCHAR(10)	NN
	paid_at	TIMESTAMP	NN

employee — D7 Staff & Stores

PK	**employee_id**	INTEGER	NN
FK	store_id	INTEGER	→ store
	first_name	VARCHAR(40)	NN
	last_name	VARCHAR(40)	NN
UK	email	VARCHAR(120)	NN
CK	role	VARCHAR(12)	NN
	password_hash	VARCHAR(255)	NN
	active_flag	BOOLEAN	NN

customer — D1 Customers & Devices

PK	**customer_id**	INTEGER	NN
	first_name	VARCHAR(40)	NN
	last_name	VARCHAR(40)	NN
IX	phone	VARCHAR(20)	NN
	email	VARCHAR(120)	
	created_at	TIMESTAMP	NN

job_part — D3 Repair Jobs

PK	**job_part_id**	INTEGER	NN
FK	repair_job_id	INTEGER	→ repair_job
FK	part_id	INTEGER	→ part
FK	purchase_order_id	INTEGER	→ purchase_order
	quantity	INTEGER	NN
CK	part_status	VARCHAR(8)	NN
	recorded_at	TIMESTAMP	NN

device — D1 Customers & Devices

PK	**device_id**	INTEGER	NN
FK	customer_id	INTEGER	→ customer
	device_type	VARCHAR(20)	
	brand	VARCHAR(40)	
	model	VARCHAR(60)	NN
IX	serial_or_imei	VARCHAR(40)	

job_status_history — D3 Repair Jobs

PK	**history_id**	BIGINT	NN
FK	repair_job_id	INTEGER	→ repair_job
CK	status	VARCHAR(20)	NN
FK	changed_by_id	INTEGER	→ employee
	changed_at	TIMESTAMP	NN
	note	VARCHAR(300)	

part — D4 Parts Inventory

PK	**part_id**	INTEGER	NN
FK	default_supplier_id	INTEGER	→ supplier
	part_name	VARCHAR(100)	NN
	device_model	VARCHAR(60)	
	grade	VARCHAR(20)	
	standard_cost	DECIMAL(10,2)	NN
	active_flag	BOOLEAN	NN

supplier — D6 Suppliers

PK	**supplier_id**	INTEGER	NN
UK	supplier_name	VARCHAR(100)	NN
	contact_name	VARCHAR(80)	
	phone	VARCHAR(20)	
	email	VARCHAR(120)	
	active_flag	BOOLEAN	NN

quote — D3 Repair Jobs

PK	**quote_id**	INTEGER	NN
FK	repair_job_id	INTEGER	→ repair_job
	labor_amount	DECIMAL(10,2)	NN
	parts_amount	DECIMAL(10,2)	NN
	total_amount	DECIMAL(10,2)	derived · NN
CK	decision	VARCHAR(10)	NN
	decision_at	TIMESTAMP	
FK	decision_by_id	INTEGER	→ employee

store_part_stock — D4 Parts Inventory

PK,FK	**store_id**	INTEGER	→ store
PK,FK	**part_id**	INTEGER	→ part
CK	quantity_on_hand	INTEGER	derived · NN
	reorder_point	INTEGER	NN

purchase_order — D5 Purchase Orders

PK	**purchase_order_id**	INTEGER	NN
FK	store_id	INTEGER	→ store
FK	supplier_id	INTEGER	→ supplier
FK	created_by_id	INTEGER	→ employee
FK	approved_by_id	INTEGER	→ employee
CK	po_status	VARCHAR(20)	NN
	created_at	TIMESTAMP	NN
	approved_at	TIMESTAMP	
	placed_at	TIMESTAMP	
	rejection_reason	VARCHAR(300)	

inventory_txn — D4 Parts Inventory

PK	**inventory_txn_id**	BIGINT	NN
FK	store_id	INTEGER	→ store_part_stock
FK	part_id	INTEGER	→ store_part_stock
FK	recorded_by_id	INTEGER	→ employee
CK	txn_type	VARCHAR(12)	NN
	quantity_change	INTEGER	NN
	reason	VARCHAR(200)	
	txn_at	TIMESTAMP	NN

po_line — D5 Purchase Orders

PK	**po_line_id**	INTEGER	NN
FK	purchase_order_id	INTEGER	→ purchase_order
FK	part_id	INTEGER	→ part
	quantity_ordered	INTEGER	NN
	quantity_received	INTEGER	NN
	unit_cost	DECIMAL(10,2)	NN

schedules / is scheduled at
books / is booked by
leads to / results from
receives / is received for
employs / is employed by
handles / is handled by
works on / is assigned to
sends notification for / has notification sent by
uses / is used on
owns / is owned by
undergoes / is performed on
tracks / belongs to
is included in / includes
creates / is created by
generates / is generated for
appears in / is stock of
changes / is made by
provides / is provided by
logs / is logged for
approves / is approved by
decides / is decided by
appears on / lists
records / is recorded by
contains / is contained in
fulfills / is fulfilled by
supplies / is supplied by

Every foreign key is drawn as a relationship line. EMPLOYEE plays several roles: the technician assigned to (or claiming) a job or appointment, the sender of the pickup notification, and the person who changes a status, decides a quote, records a stock change, or creates or approves (a Manager) a purchase order.

QuickPhone Repair Tracking System: Physical ERD notes

MIS 4173 Sprint 3 · Team 4 · Method: Dennis, Wixom, Roth & Haley, *Systems Analysis and Design*, 8th ed., Ch. 5 and Ch. 10

Logical to physical (Fig. 10-7)

1. **Entities to tables.** Sixteen tables in snake_case; each header names the DFD data store (D1–D7) that holds it, so the ERD and the DFDs stay balanced.
2. **Attributes to fields.** Each field has a data type and size (VARCHAR, INTEGER, DECIMAL, TIMESTAMP, BOOLEAN), NN for not null, and CK where only listed values are allowed. Defaults and valid values (Fig. 10-9) are in the models document, Appendix A.
3. **Primary keys.** Every table has one (PK); store_part_stock uses the pair store_id + part_id.
4. **Foreign keys.** Every relationship is a foreign key in the child table (FK, with the parent table named). Every foreign key is drawn as a line with crow's-foot cardinality, and each line is named both ways, as in Fig. 10-8 ("Places / Is Placed By"): the first name reads from the "one" side to the "many" side, the second reads back (Ch. 5).
5. **System-related components.** Each field the physical DFD adds is stored where its flow writes it: password_hash by 9.1 Maintain Staff Accounts (Staff Account Input Form); ticket_number by 2.4 Create Repair Job, printed by 2.5 on the Claim Ticket and Device Tag Printouts; email_sent_at, email_sent_by_id and email_status by 5.5 Record Notification Result (Email Send Result); id_verified_flag by 6.2 Record Identity Check (ID Check Input Form); signature_file and picked_up_at by 6.4 Close Repair Job (Signature Input Form); device_passcode, encrypted, by 2.4 (Passcode Input Form); assigned_tech_id by 2.4 or by a claim at 3.1 (Job Claim Input Form). The audit tables take a row from every status change (job_status_history) and from 4.3, 7.5, 9.3 and the 9.5 Inventory Import File batch (inventory_txn); time stamps are set on save. The Import Error Report is not stored.

Documented denormalizations

The logical ERD is in third normal form. Three fields marked *derived* are stored on purpose for speed and are kept correct by the program (Ch. 10, optimizing storage):

- `repair_job.status`: the latest row of job_status_history.
- `quote.total_amount`: labor_amount + parts_amount.
- `store_part_stock.quantity_on_hand`: the sum of inventory_txn.quantity_change.

Assumptions

Types are standard SQL, all valid in PostgreSQL, the chosen DBMS. DEVICE stays its own table inside D1: storage is optimized table by table (no repeated customer data, few nulls; Ch. 10), so a separate data store would save nothing. EMPLOYEE.store_id is a home store; the Owner sees both stores. Photo ID numbers are never stored, only that the check was done.

Owner clarifications (October 23)

device_type and brand are optional (intake records name, phone, model and issue). assigned_tech_id and appointment.technician_id stay empty until a job is assigned or claimed. device_passcode is optional, encrypted, for authorized staff only (proposed: assigned technician, Manager, Owner). approved_by_id names a Manager. An unrepaired job has no payment row (no diagnostic fee). The Manager sets reorder_point; a part request is a job_part row with part_status Needed.